



Reference Cluster Design Architecture MI300X / MI325X

128 GPU to 1024 GPU

Version 4.2 – March 2026

Cluster - 128 GPU to 1024 GPU (16 to 128 Nodes) *

Cluster Size	128 to 1024 GPU
Platforms	Dell XE9680 Lenovo SR685a V3 SMCI AS-8125GS
OS	Ubuntu 22.04 (or above)
Linux Kernel	5.15 – 6.80
ROCm	6.3.3 (or above)

*Genericized BoM – removing device specifics including dependencies

Storage Type	
Local Storage	1.6 TB (or greater)
Utility Storage	Pure, Vast, RYO
Bulk Storage	Pure, Vast, WekaIO
Scratch Storage	Vast, DDN, WekaIO, Hammerspace
Archive/Object Storage	S3 Compatible

Network - 128 GPU to 1024 GPU (128 to 1024 Nodes)*

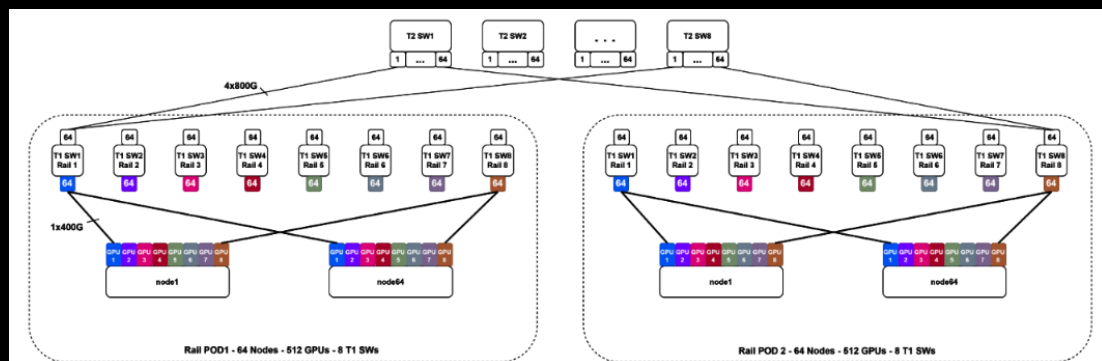
Backside Network Topology	2 Tier Rail Optimized / Fat Tree
NIC	Pollara 400 BCM957608 (Thor2)
Switch	Arista, Dell, Juniper, Cisco (TH 4/5, Scheduled fabrics), Nokia
Network OS	SONiC, Junos, EOS, IOS, DriveNets
Subscription Ratio	1:1.16=16% Undersubscribed (AMD recommendation)
Optics	Vendor ACL/HCL transceivers or Direct Attach Copper
Fabric	RoCEv2 Ethernet

Frontside Network Segment	Adapter Recommended
All-in One Network	Ethernet 100GbE 2-port QSFP28 Adapter
Storage Network (Optional)	Ethernet 100GbE 2-port QSFP28 Adapter
Virtualization Network (Optional)	Ethernet 100GbE 2-port QSFP28 Adapter
Host In-Band	Ethernet 10/25GbE 4-Port SFP28 Adapter
BMC OOB Mgt	1G Copper

* Genericized BoM – removing device specifics including dependencies

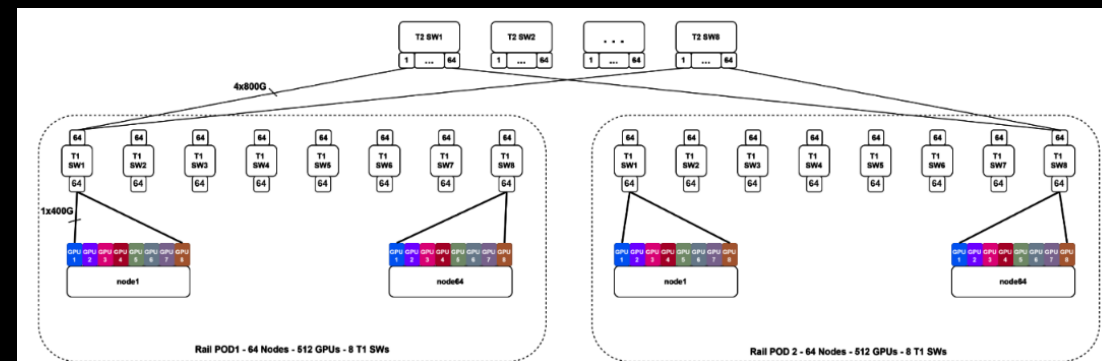
128 GPU to 1024 GPU – General Network Layouts

Rail Optimized (1:1 example)



- Rail design - 2 pods
- 64 Nodes per pod – 512 GPUs
- Sizing of network can be done through lower/increase spine layer and adjusting pod size.
- Lower latency for AI workloads – optimized CPU communication
- Improved performance for large scale models

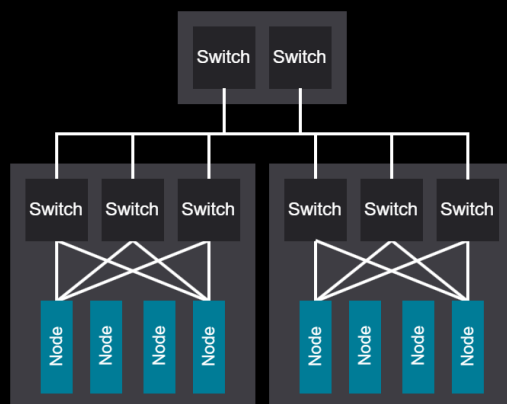
Tree Topology (1:1 example)



- Tree design – 2 pods
- 64 Nodes per pod – 512 GPUs
- Sizing of network can be done through lower/increase spine layer and adjusting pod size.
- Non blocking bandwidth and scalability
- Load balancing and fault tolerance
- Diverse workloads

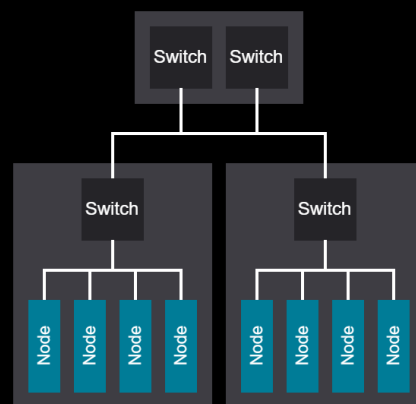
Cluster Network

Basic Network Topologies



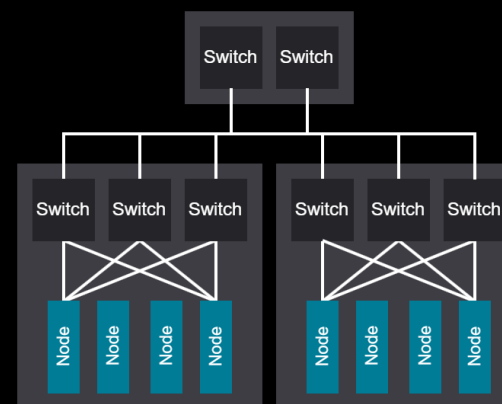
2-Tier Rail Network

- Enables large scalable unit sizes for large jobs or replica sizes
- Efficient for workloads favoring ring-based collectives
- Higher infrastructure costs



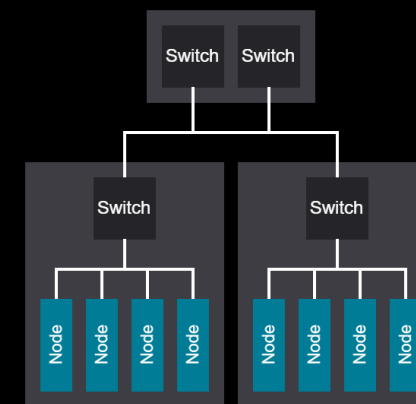
2-Tier Tree Network

- Efficient for small workloads or replicas
- Easy to add capacity with proper planning
- Potentially allows for lower infrastructure costs
- Limited blast radius compared to rail networks



3-Tier Rail TH5/J3 Network

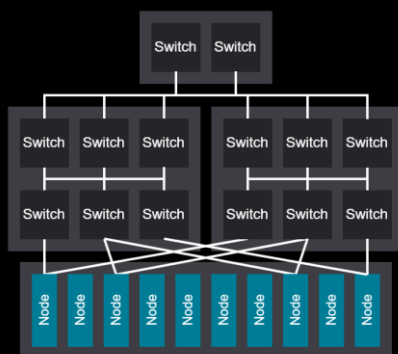
- Spine switches replaced with a 2-tier Jericho3-AI/Ramon3 fabric for increased maximum cluster size
- Deep buffers and scheduled fabric aid greatly with congestion issues in large clusters at a small latency penalty



3-Tier Tree TH5/J3 Network

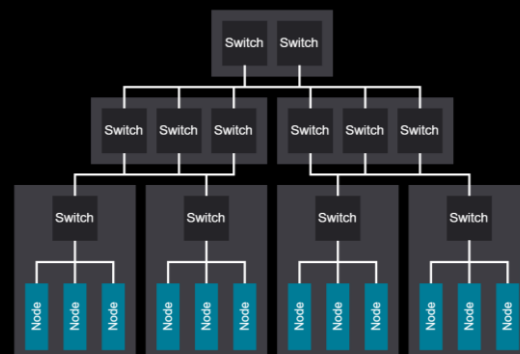
- Same benefits from switch to scheduled spine fabric as with rail, but retains the primary characteristics of tree networks

Basic Network Topologies Continued



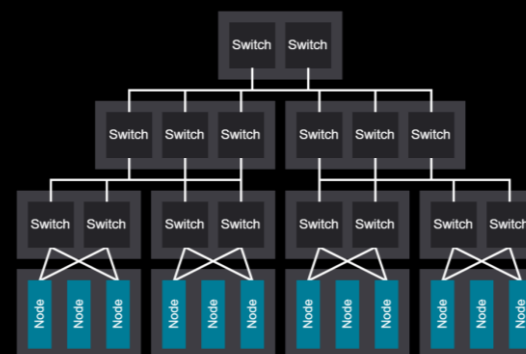
3-Tier Rail Optimized Network

- Allows for massive scalable unit sizes
- Best ring-based collective performance at scale; at the cost of poor any-any performance



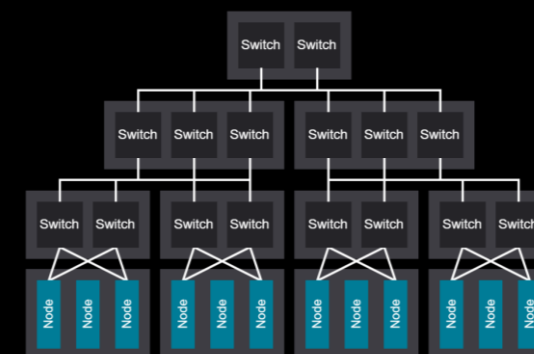
3-Tier Tree Network

- Allows for massive cluster sizes
- Best any-any performance at scale
- Suitable for a “campus style” deployment



3-Tier Hybrid Rail Network

- Allows for massive cluster sizes with large scalable units
- Favors ring-based collectives, but does not sacrifice significant any-any performance on large jobs
- Suitable for a “campus-style” deployment

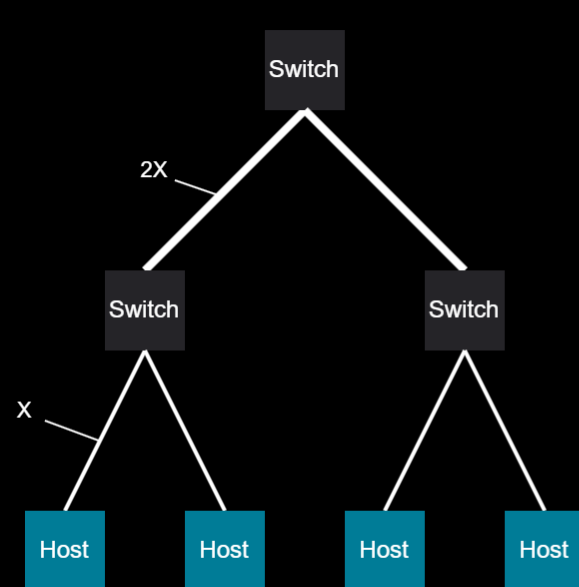


3-Tier Fully Scheduled Rail Network

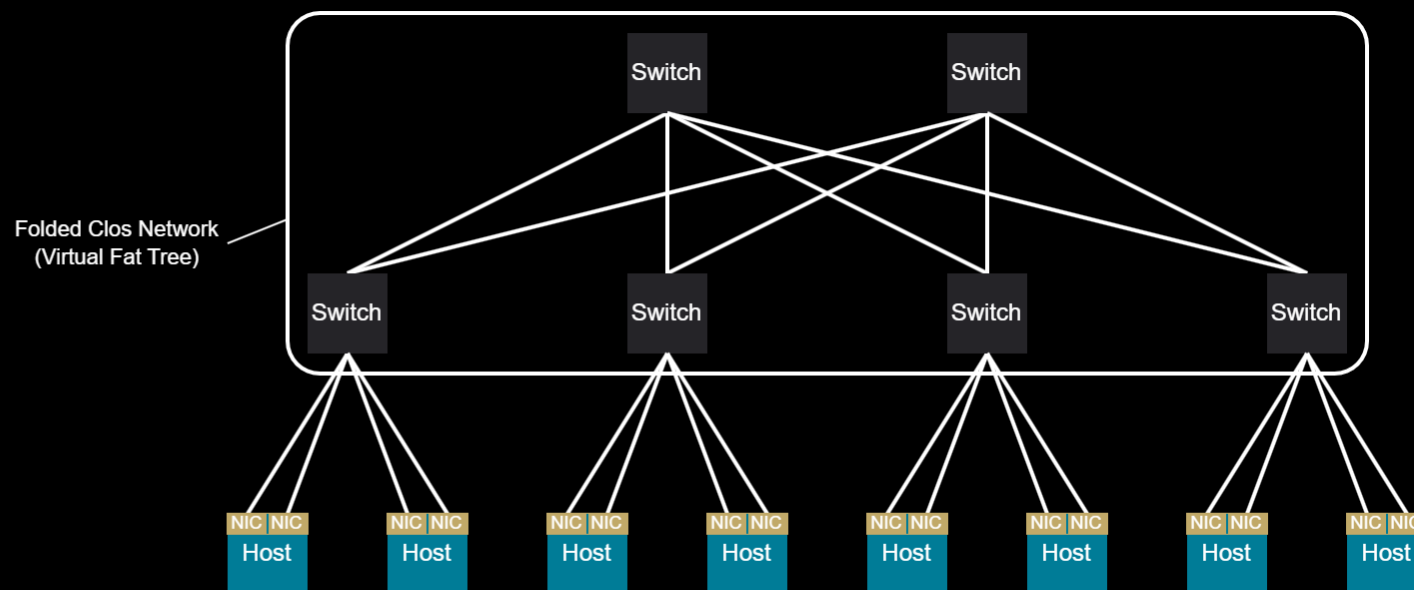
- Medium sized scalable units
- Excellent congestion performance due to deep buffers and scheduled fabric
- Technical limitations limit cluster size to ~32K GPUs in recommended configuration

Tree and Rail

Fat Tree Terminology In Cluster Design



Fat Tree Topology

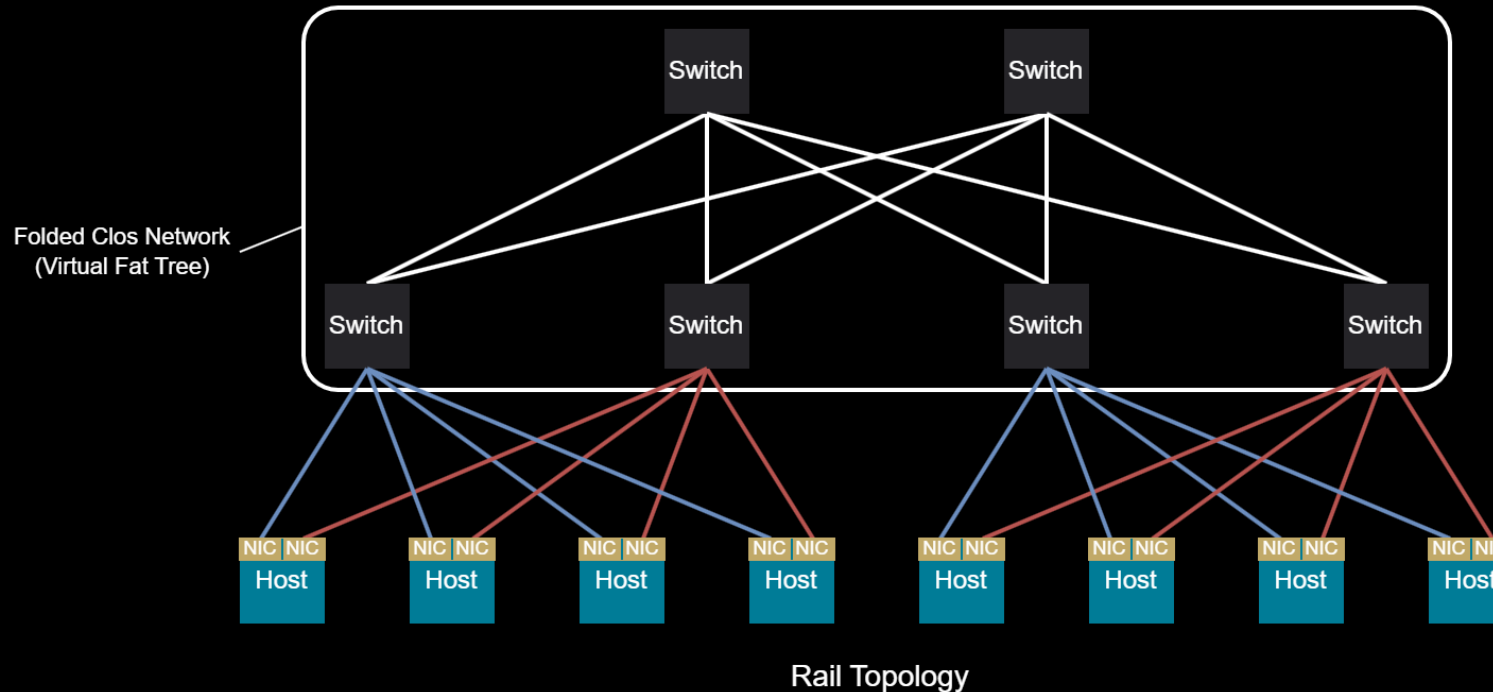


"Fat Tree" Topology

The canonical fat tree topology is a network concept where a switch's connection to upstream peers has at least parity bandwidth with the total aggregate bandwidth of its downstream connections. This causes links between switches to become “fatter” as they get closer to the core.

The “fat tree” topology for AI/ML clusters instead refers to how a host is connected to its upstream switches; in this case all host NICs terminate on the same switch. It can also be considered 1-rail network. The network itself is generally a 3-stage or 5-stage folded Clos network due the fixed radix of network switches.

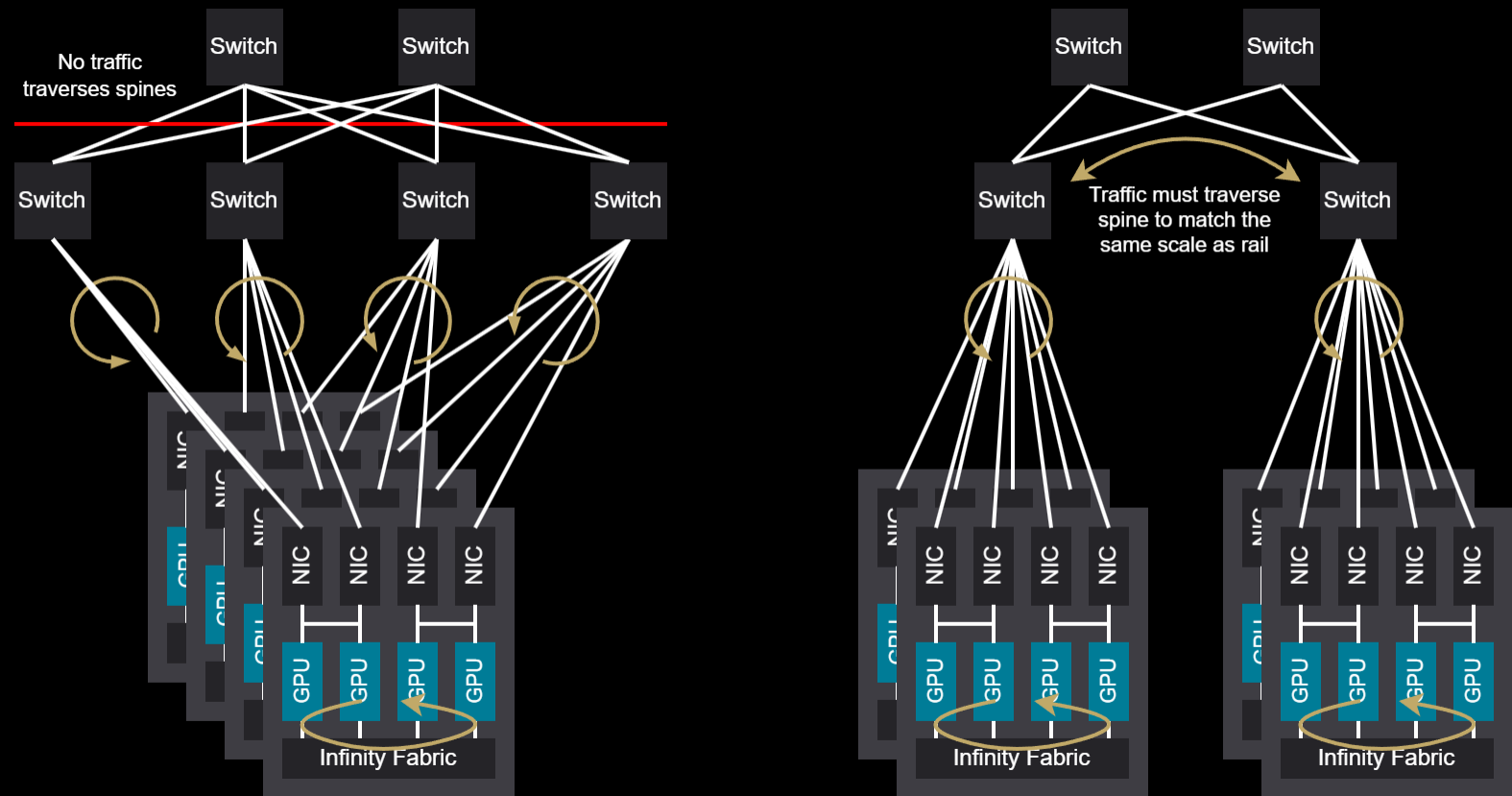
Rail Networks In Cluster Design



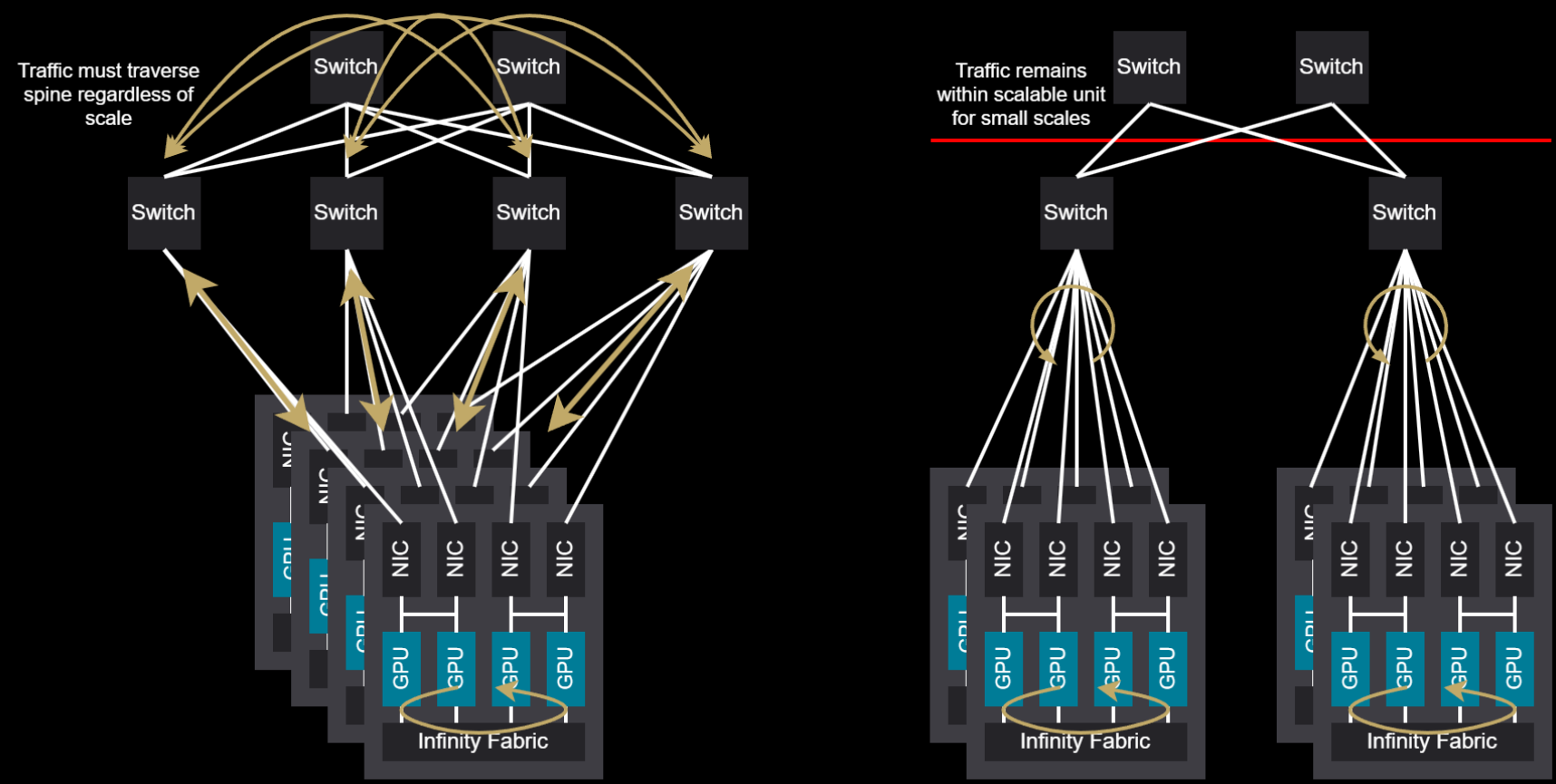
Rail networks leverage the same folded Clos network as tree networks, but host connections are instead aggregated onto switches based on NIC rank. These shared ranks are referred to as rails and allow the network to provide preferential latency for connections which share the same rail. The downside to this design is any traffic which needs to cross rails/ranks must traverse either the network spine layer, or Infinity Fabric (PXN).

The above example shows an example 2-rail network, with the rails colored blue and red to differentiate them.

Rail Enables Larger Single Hop Ring Domains

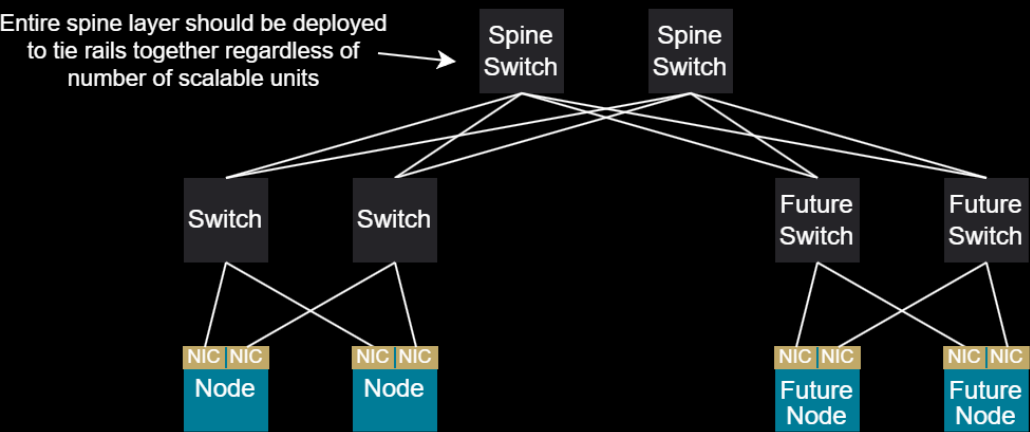
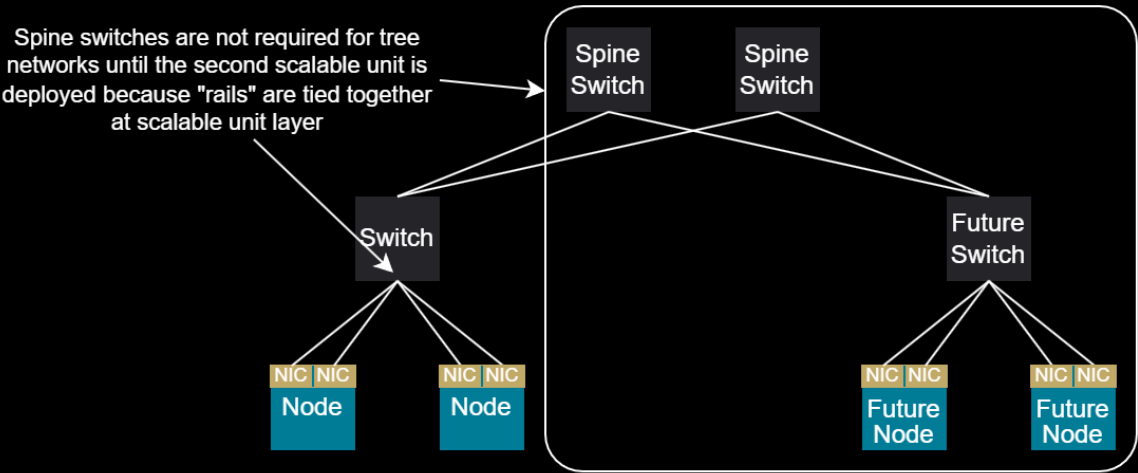


Tree Handles Cross-Rank Traffic Better

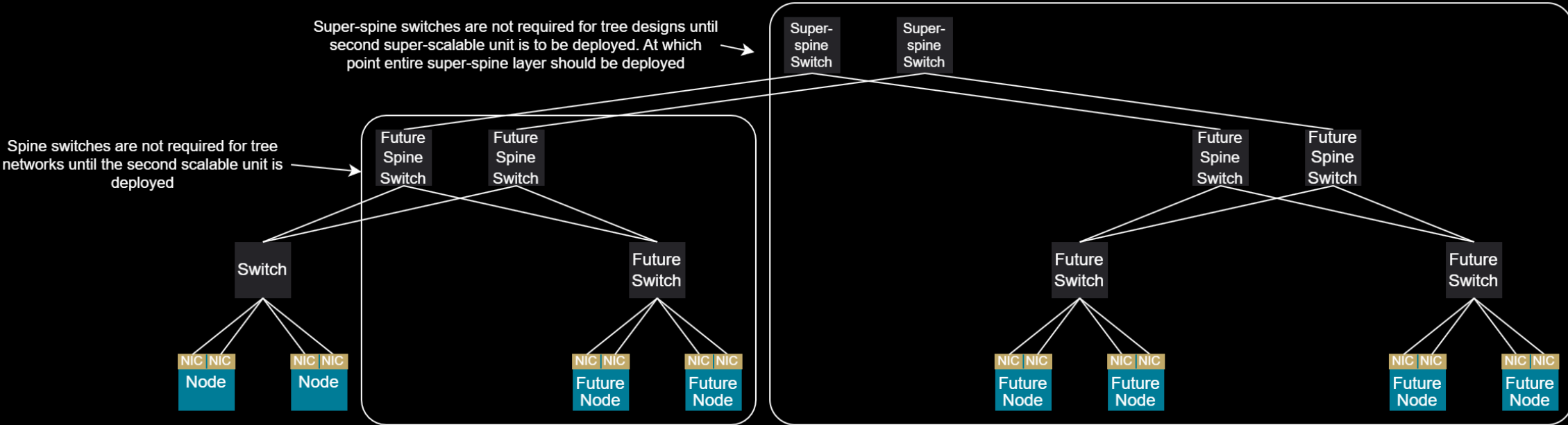


Scaling Networks

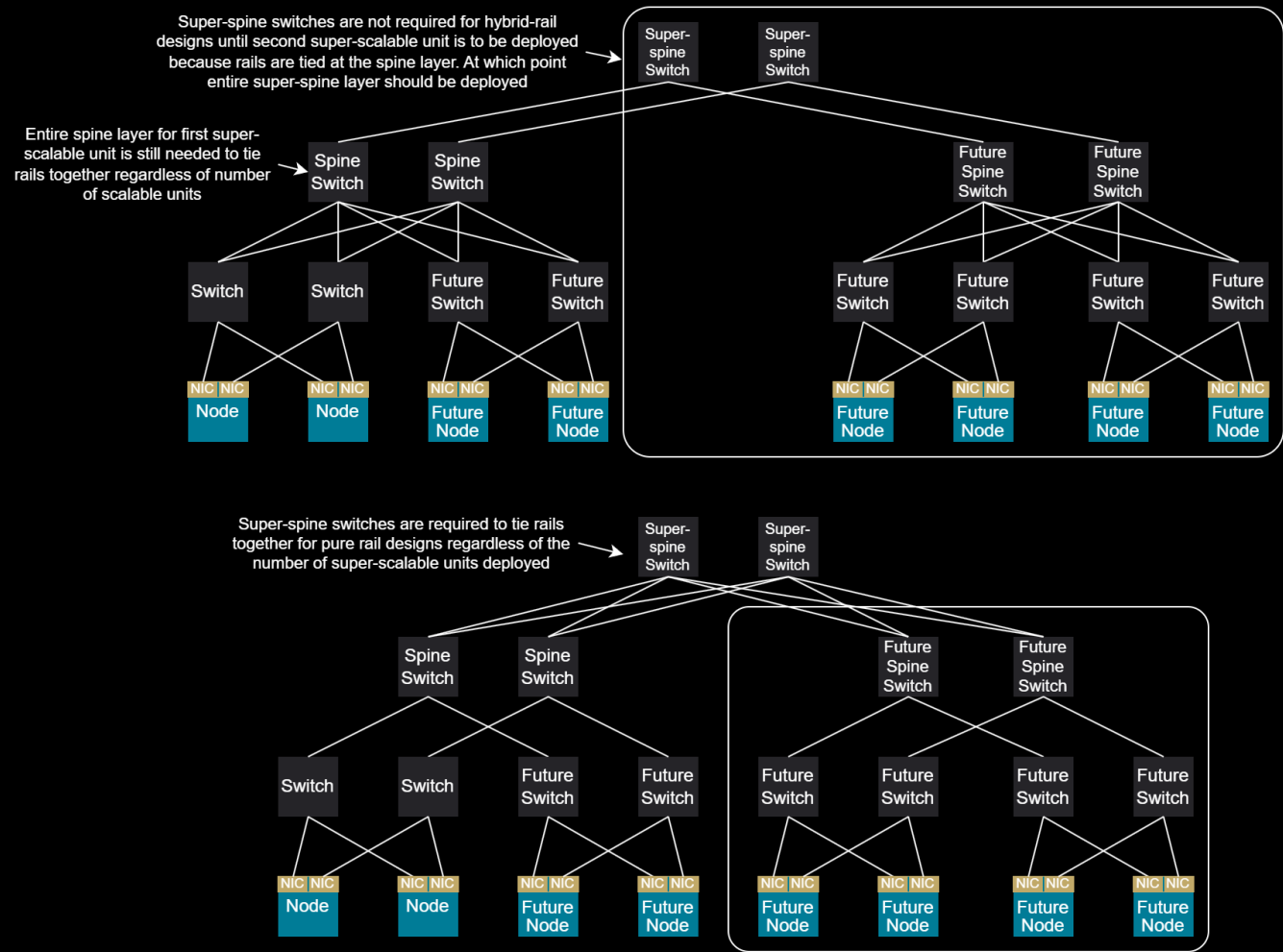
Cluster Backend Deployment Strategy for 2-tier Networks



Cluster Backend Deployment Strategy for 3-tier Tree Networks



Cluster Backend Deployment Strategy for 3-tier Rail Networks



Network Subscription

Subscription is the relationship between what is provided by the upstream network and what is required by the downstream network in demand side.

It is typically represented as a ratio:

$$\text{Downstream Demand} : \text{Upstream Capacity}$$

1:1 subscribed network would be equal downstream capacity to upstream capacity.

Example: 1:1.16 = (1 downstream demand : 1.16 upstream capacity. In this example there is .16 more upstream capacity)

Or as a percentage:

$$\text{Subscription Rate} = \frac{\text{Downstream Demand}}{\text{Upstream Capacity}}$$

80% subscription ratio could be referred to as “20% undersubscribed”, or a 120% subscription ratio could be referred to as “20% oversubscribed”.

Network Design Examples

Network Topology Design Notations:

Design note:

Designs included are based on scheduled fabrics switch type (Arista, Ciena, Nokia) or 51.2T switch type (Arista, Cisco, Dell, Juniper)

Vendors/switch models vary for port count and features – please consult desired vendor port count directly to confirm.

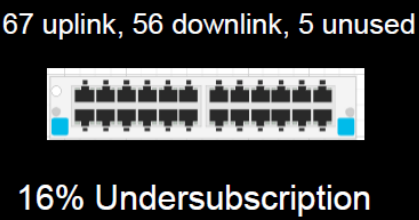
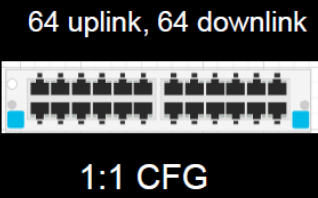
Scalable Units/PODs note:

Diagrams presented are designed around a Scalable Unit or POD – which can determine overall network end to end latency and AI use cases.

Certain ML/AI workloads may require change of scalable unit size. Please consult with AMD Architecture as required.

Network Layout - 128 GPU Cluster 16 Nodes – Single Switch

Example 128 port switch



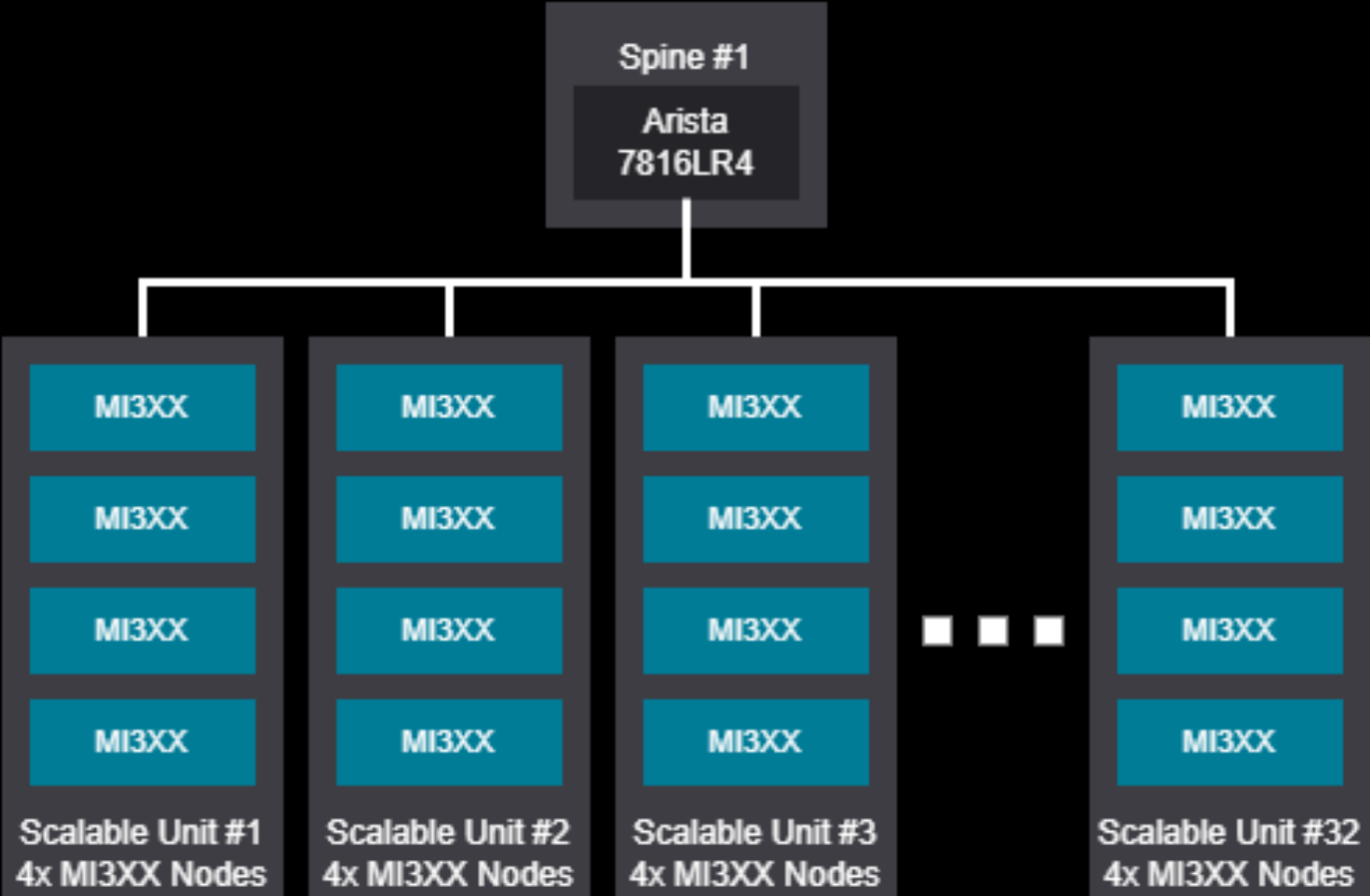
Fat Tree / Rail

Tier	Leaf switch	Spine switch	Superspine switch
3 tier	16%	1-1	All

Network Diagram - 128 GPU to 1024GPU (16 to 128 Nodes)

Tree Design

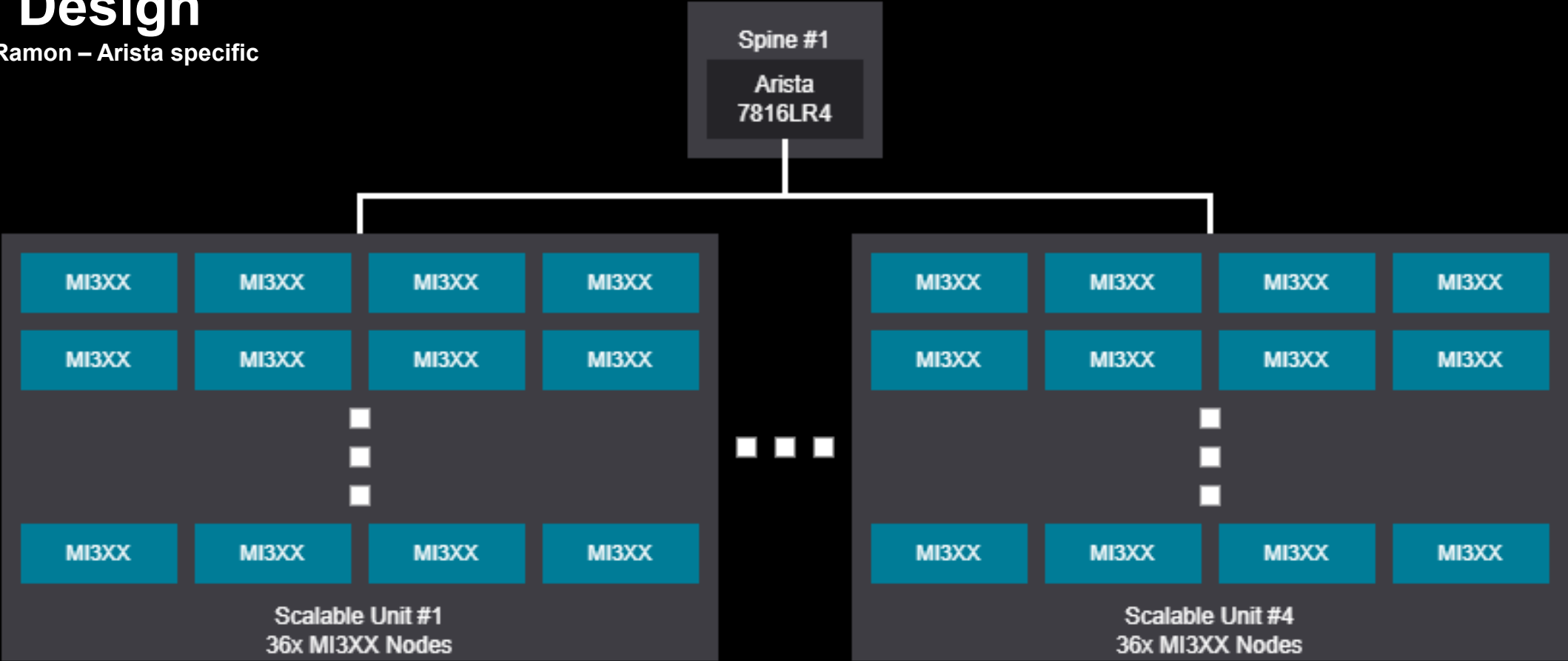
Jericho/Ramon – Arista specific



Network Diagram - 128 GPU to 1152GPU (16 to 144 Nodes)

Rail Design

Jericho/Ramon – Arista specific

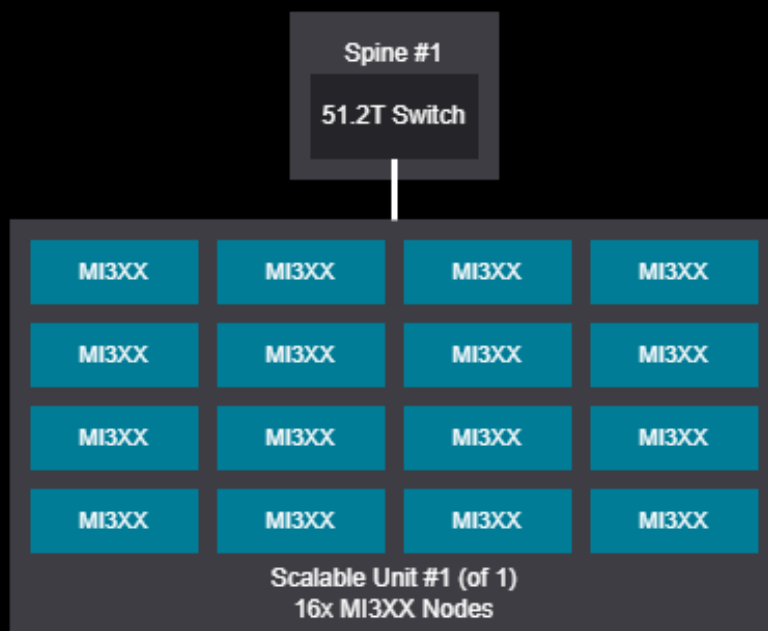




Appendix 1 - Topological Variations

128 GPU Generic Topology Design Examples

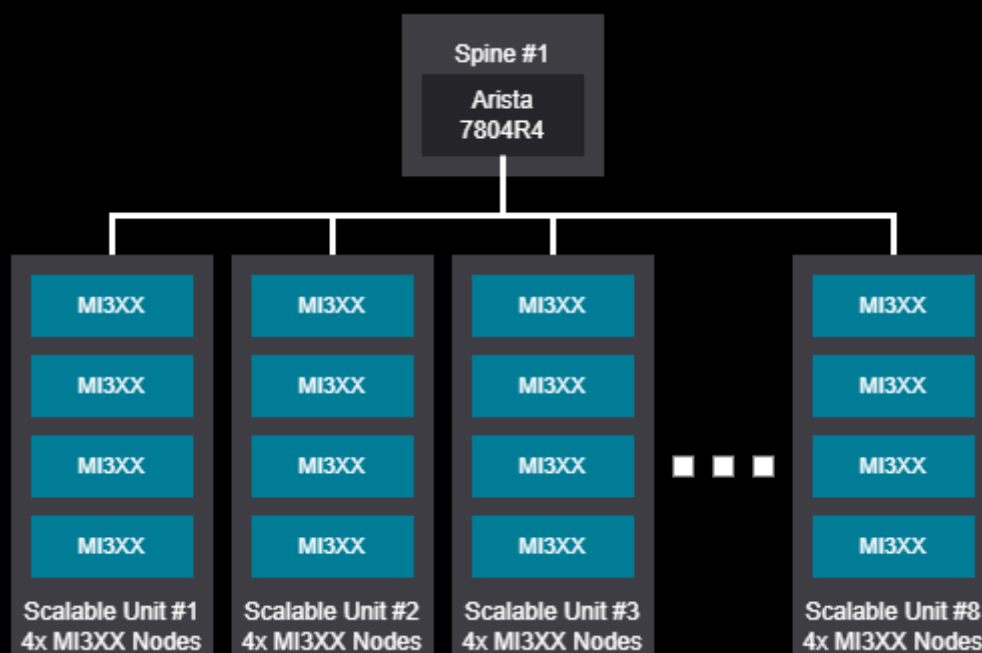
8-128 GPU Single Switch Design



- Can use Cisco G200 or Broadcom Tomahawk 5
- Single switch design will give the best performance possible compared to any alternative
- Extremely simple configuration and deployment

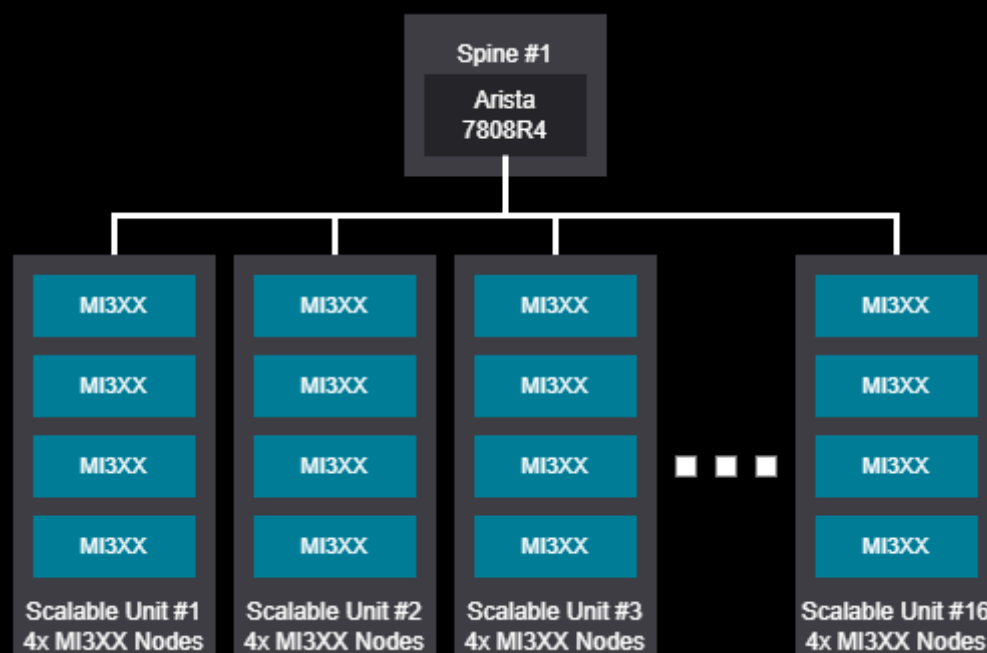
129-1024 GPU Tree Designs – Scheduled Fabric Specific

129-256 GPU Tree Design



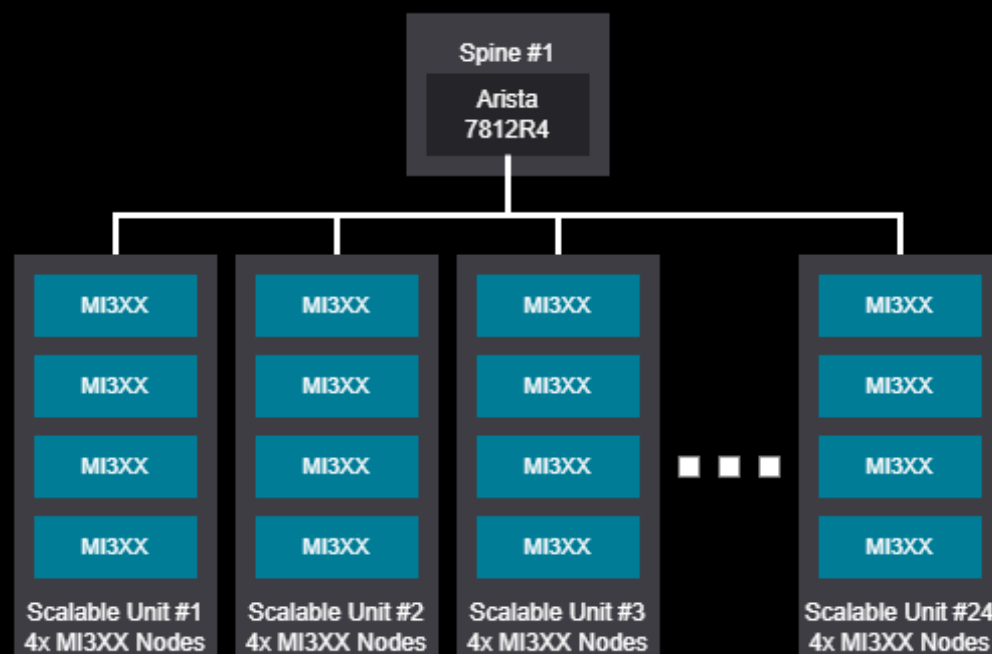
- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Tree can scale to a slightly smaller cluster than rail

257-512 GPU Tree Design



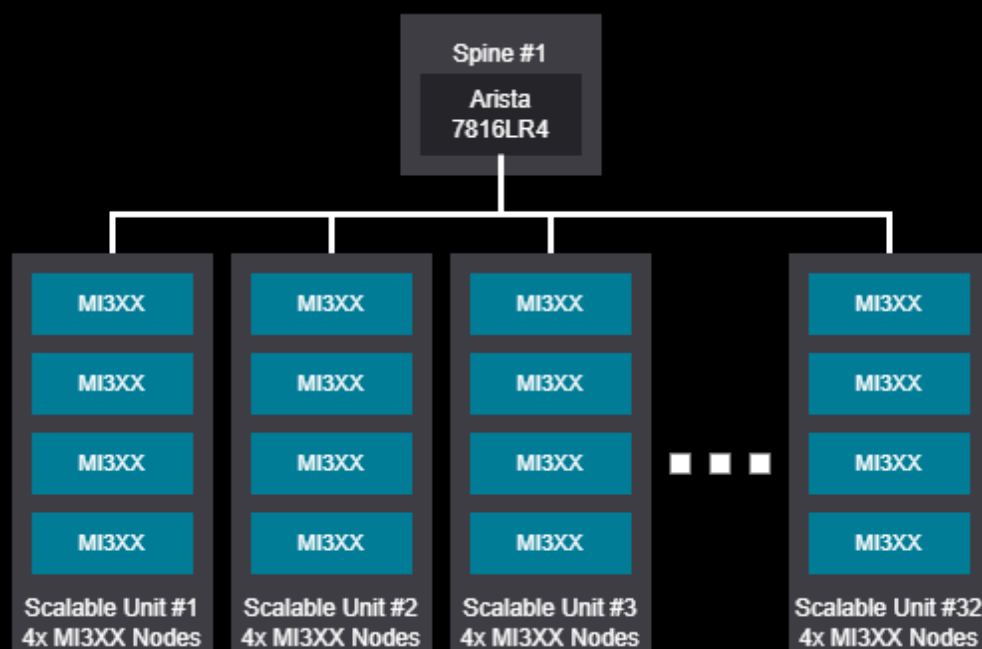
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513-768 GPU Tree Design



- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
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- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Tree can scale to a slightly smaller cluster than rail

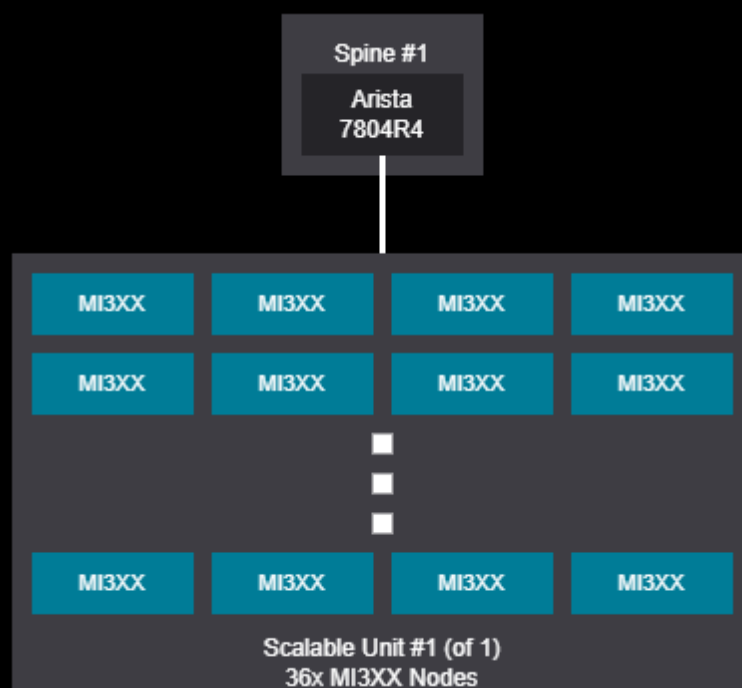
769-1024 GPU Tree Design



- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Tree can scale to a slightly smaller cluster than rail

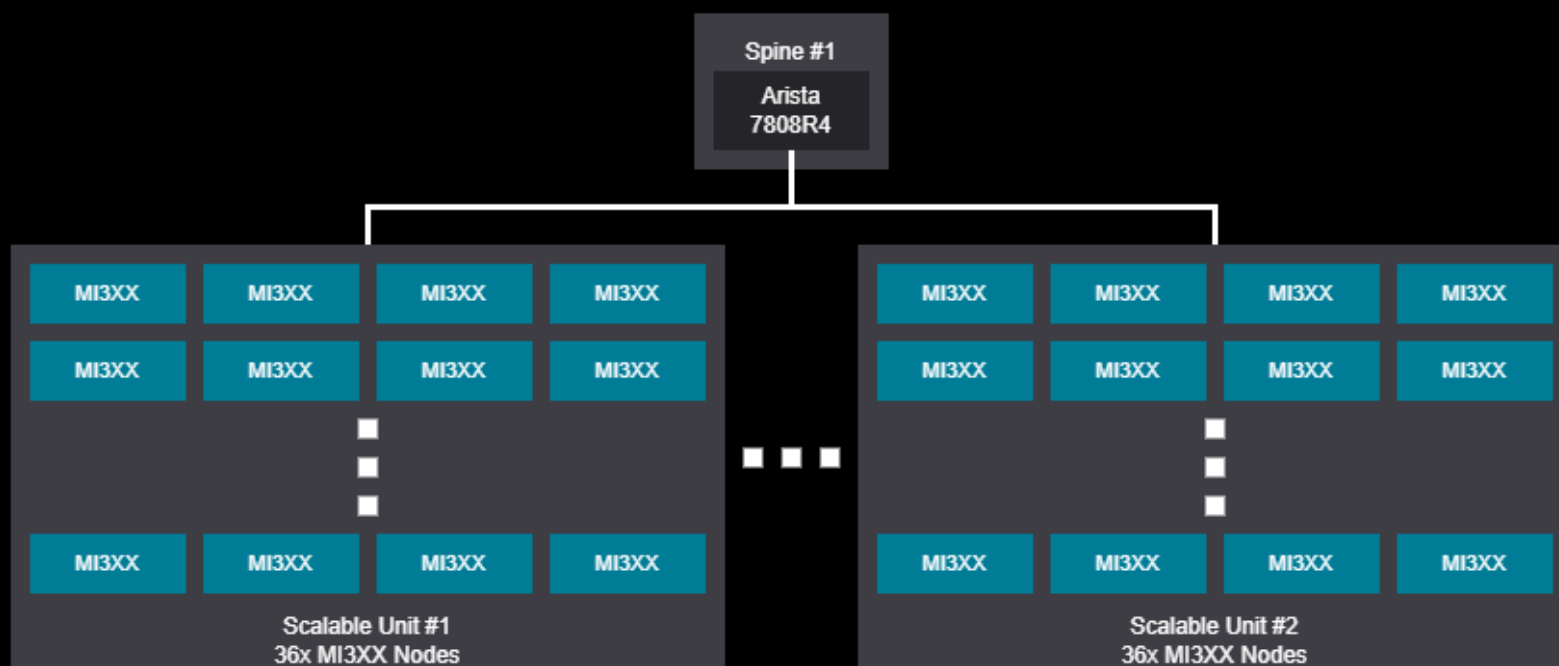
129-1152 GPU Rail Designs – Scheduled Fabric Specific

129-288 GPU Rail Design



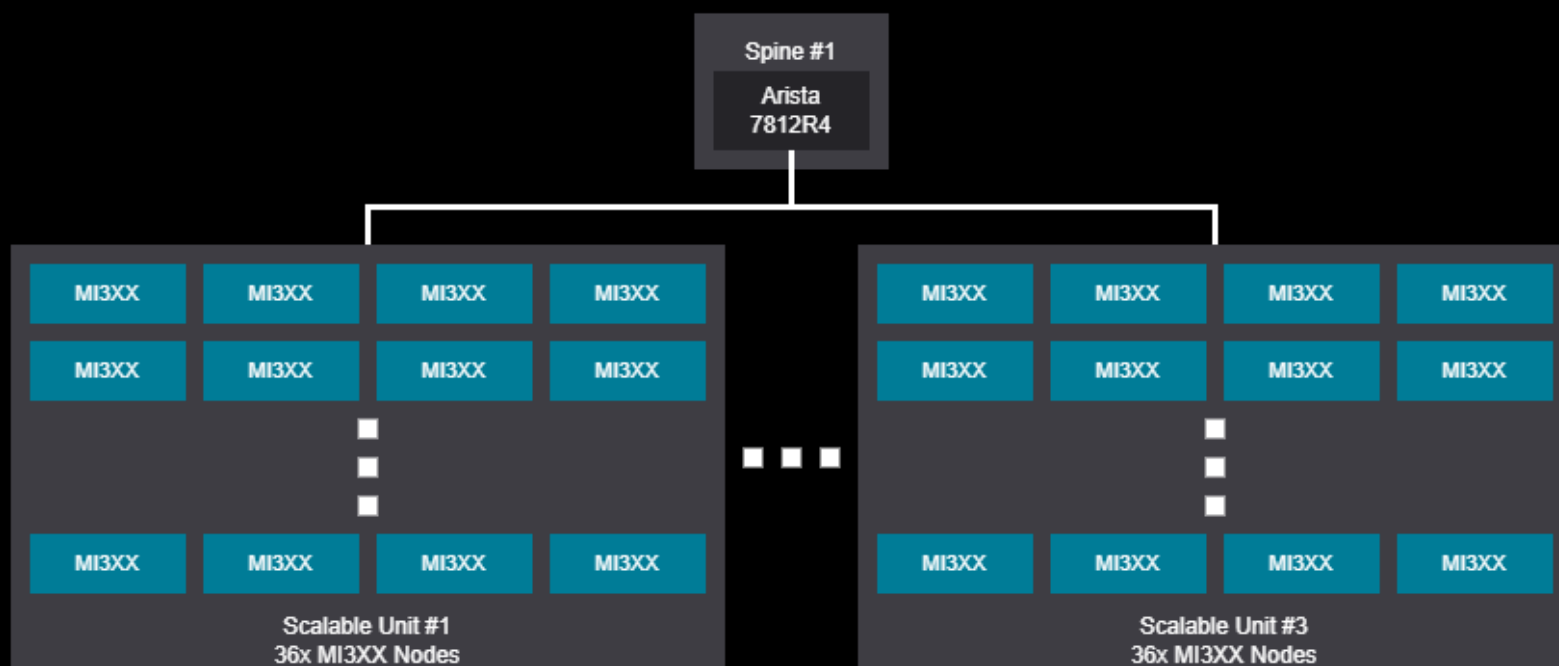
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- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Rail can scale to a slightly larger cluster than tree

289-576 GPU Rail Design



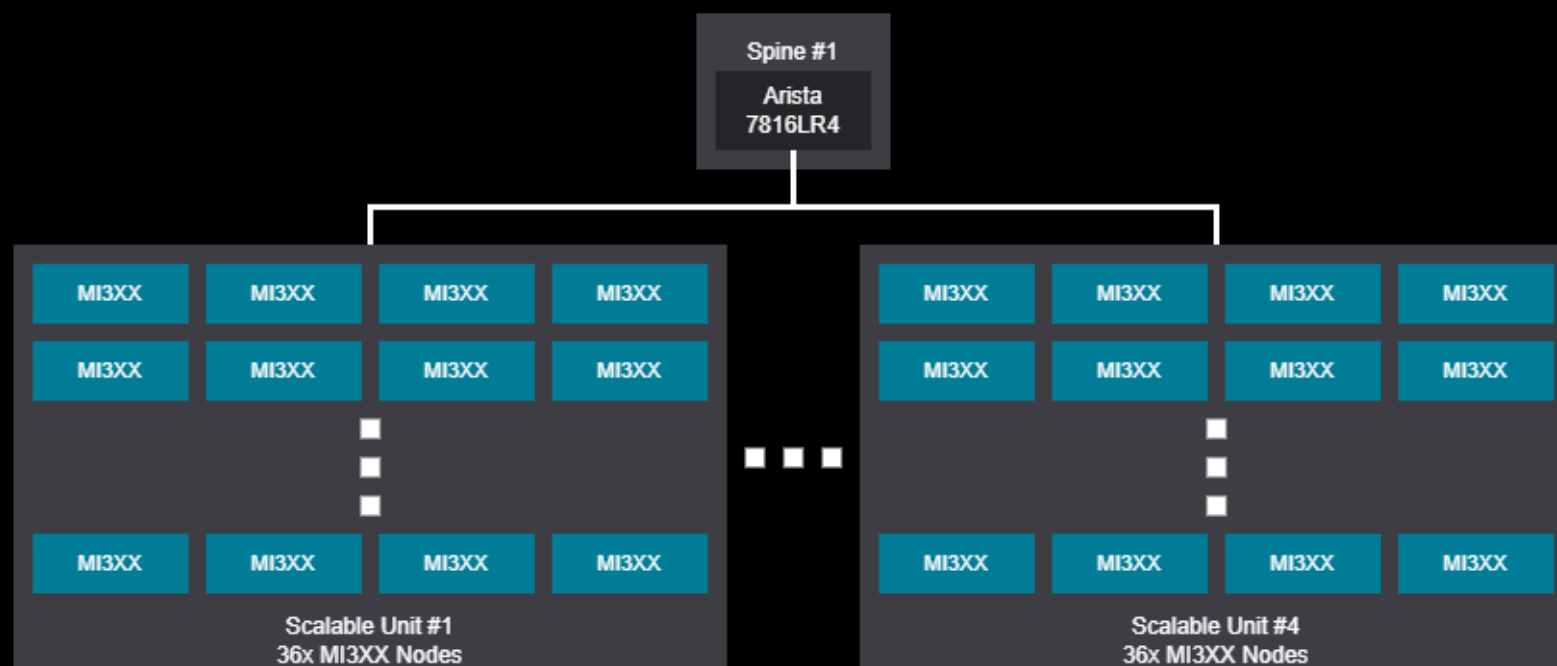
- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Rail can scale to a slightly larger cluster than tree

577-864 GPU Rail Design



- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Rail can scale to a slightly larger cluster than tree

865-1152 GPU Rail Design



- Built around the Arista 7800R4 Jericho3-AI/Ramon3 series chassis switch
- Functionally a 2-tier network, but managed as a single switch
- Fully scheduled fabric
- Extremely simple configuration and deployment
- Tree vs rail topology is accomplished by adjusting which ports a node is connected to the switch
- Rail can scale to a slightly larger cluster than tree

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